

Session Objectives

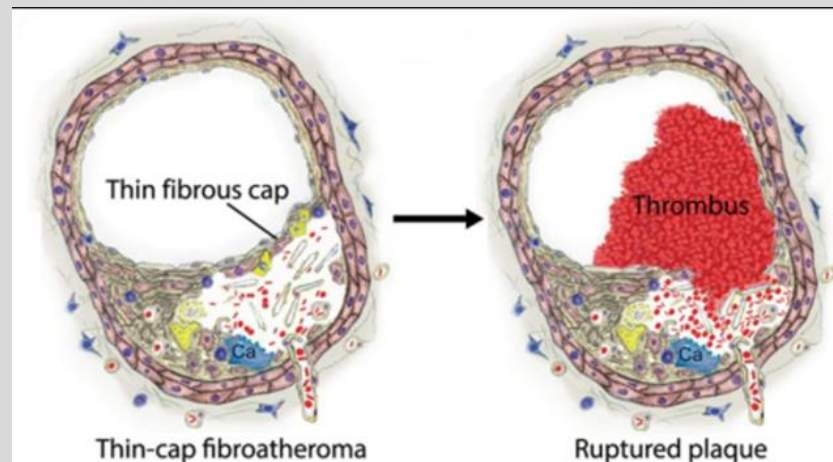
- 1. Describe and discuss the general concepts of **ischemic heart disease (including the epidemiology and risk factors)** and **define and compare stable, unstable, and Prinzmetal angina (review)**.
- 2. Explain the **pathogenesis/mechanism of coronary thrombosis and how it leads to myocardial infarction** and **delineate the initial changes of myocardial necrosis at the cellular and subcellular levels**.
- 3. Discuss, describe, compare, and differentiate the **changes of myocardial infarct as the time sequence evolves in morphological and pathophysiological terms**.
- 4. Correlate **myocardial infarction changes with clinical and laboratory manifestations**.
- 5. Discuss, describe, and evaluate **complications of myocardial infarctions including arrhythmias, pericarditis, ventricular free wall rupture, true and false aneurysms, papillary muscle and septal rupture, and contractile dysfunction**.

IHD – Consequences of Myocardial Ischemia

- Stable angina – Previously discussed, chest pain with physical activity
- Unstable angina – Previously discussed, chest pain at rest
- Prinzmetal angina – vasospasm- does not depend on exercise or rest
- Myocardial infarction – acute plaque change with total thrombotic occlusion, followed by myocardial death
- Sudden cardiac death – often due to a disrupted plaque producing ischemia which induces a fatal ventricular arrhythmia

IHD - Pathogenesis

- Over 90% of patients with IHD have atherosclerosis within one or more coronary arteries
- Obstructions occupying 75% or more of the lumen cause symptoms induced by exertion
- 90% obstruction can lead to insufficient coronary blood flow even at rest
- Acute coronary syndrome
 - **#1 source of MI is acute plaque rupture of vulnerable or unstable plaque with thrombus formation**
 - Occurs via rupture, erosion, ulceration, and/or deep hemorrhage
 - Often associated with intralesional inflammation, which mediates plaque disruption



Myocardial Infarction

- Necrosis of cardiac myocytes (Irreversible)
- Usually due to rupture of an atherosclerotic plaque with thrombus - complete occlusion of coronary artery
- Leading cause of death in U.S

Myocardial Infarction

- Transmural MI =
 - ischemic necrosis involves full thickness of ventricle wall (in distribution of single coronary artery)
 - most MIs are transmural
 - associated with coronary atherosclerosis, acute plaque change, superimposed thrombosis
- Subendocardial infarct
 - area of ischemic necrosis limited to inner one third or less than half of ventricle wall
 - (n.b. the subendocardial zone is the least well-perfused area of the myocardium - most vulnerable to reduction in coronary blood flow)

Clinical features

- Severe, crushing chest pain lasting greater than 20 minutes
- Radiation to neck jaws, arms, back commonly left side
- Other symptoms include diaphoresis, dyspnea, weakness or fatigue, nausea and vomiting, sense of impending doom, syncope
- Symptoms not relieved by nitroglycerin
- Silent MI's may occur in individuals with diabetes and elderly due to pain threshold or problems with the nervous system

Frequency and Sites of Myocardial Infarction

- Occlusion **Left Anterior Descending Artery (LAD)**: infarction of LV anterior wall, anterior septum and apex circumferentially
 - *LAD is most commonly involved artery in MI (40-50%)*
- Occlusion **Right Coronary Artery (RCA)**: infarction of posterior wall, posterior septum, and papillary muscles of LV
 - *RCA is 2nd most commonly involved artery in MI (30-40%)*
- Occlusion: **Left Circumflex Artery**: infarction of lateral wall of LV except apex (15-20%)

MI Diagnosis

- Clinical: Symptoms
- EKG/ECG changes
- Biochemical markers: myoglobin, troponins T and I, creatine kinase (CK-MB), lactate dehydrogenase
- Troponins: proteins that regulate Ca-mediated contraction of cardiac and skeletal muscle.
 - **Tn-I and Tn-T are cardiac specific** and rise 2 to 4 hours after an acute MI (up to 12 hours)
 - peak at 24 - 48 hours
 - **remain elevated for 7 to 10 days** Troponins elevated longer
- **CK-MB** also rises within 4 hours (up to 8) of an acute MI
 - peaks at 24 hours
 - **returns to normal by 72 hours**
- CK-MB is useful for detecting re-infarction that occurs days after an initial MI

Review of Cellular Injury

- Cellular injury: stress exceeds cell's ability to adapt
- Hypoxia or low oxygen delivery to tissue = cellular injury
- Causes of Hypoxia include
 - Ischemia – decrease blood flow through an organ
 - Hypoxemia – low partial pressure of oxygen in blood
 - Decreased oxygen carrying capacity – Hemoglobin loss or anemia

MI, continued

- Early biochemical consequence myocardial ischemia
end of aerobic glycolysis within seconds
 - accumulation of noxious breakdown productions (e.g. lactic acid because of increase in anaerobic glycolysis)
 - loss of contractility occurs within 60 seconds of ischemia
- Early changes are still reversible (EM changes)
 - it takes at least 20 to 40 minutes for irreversible damage (necrosis) to cardiac myocytes.
 - What does this imply clinically?

Myocardial Infarction Changes

TIME	GROSS	MICRO
0 – 30 minutes; Reversible	None	None
30 min – 4 hours	None	Early “ wavy fibers ”
4 – 12 hours	Focal mottling	Start of coagulative necrosis , edema, hemorrhage
12 – 24 hours	Dark mottling	Coagulative necrosis , pyknosis of nuclei, myocyte hypereosinophilia, contraction band necrosis, start of neutrophil infiltrate
1 – 3 days	Mottling with yellow-tan infarct center	coagulative necrosis with loss of nuclei and striations, interstitial infiltration of neutrophils

Myocardial Infarction Changes

TIME	GROSS	MICRO
3 – 7 days	Hyperemic border, central yellow-tan softening	Start of disintegration of dead myofibers, early phagocytosis of dead cells by macrophages
7 – 10 days	Maximal yellow-tan, soft with depressed red-tan margins	Phagocytosis well under way; early granulation tissue
10 – 14 days	Red-grey depressed infarct borders	Granulation tissue well formed with angiogenesis, collagen
2 – 8 weeks	Grey-white scar, from border to center of infarct	Increased collagen , decreased cellularity/vascularity
> 2 months	Scarring complete	Dense collagen scar

Robbins, Pathologic Basis of Disease, 10th edition, 2020, pg. 547, Table 12-5

Complications of MI (gross and microscopic)

- Cardiogenic shock (massive infarction), Congestive heart failure (Low EF)
- **Arrhythmia** (conduction disturbances and myocardial irritability)
- **Acute fibrinous Pericarditis** presents as chest pain with friction rub
- **Rupture** of Ventricular free wall leads to cardiac tamponade
- **Rupture** of Septum (leads to LV->RV shunt)
- **Rupture of papillary muscle** – severe, acute mitral regurgitation
- **Ventricular Aneurysm - Mural thrombus: combo of local myocardial contractility abnormality (causing stasis) with endocardial damage (thrombogenic surface)**
- **Dressler syndrome** - autoimmune pericarditis

Question

- A 55 year old man has sudden onset of severe substernal chest pain that radiates to the neck. On PE, he is afebrile but has tachycardia, hyperventilation, and hypotension. Auscultation reveals no cardiac murmurs. The patient undergoes emergent coronary angiography which shows a thrombotic occlusion of the left circumflex artery and 50 to 70 % narrowing in the proximal circumflex and anterior descending arteries. Which of the following complications of this disease is most likely to occur within one hour of these events?
- A. ventricular fibrillation
- B. Pericarditis
- C. Myocardial rupture
- D. Ventricular aneurysm
- E. Thromboembolism

Pericarditis Clinical and Lab Findings

- Pericardial effusion- Muffled heart sounds
- Hypotension associated with pulsus paradoxus(abnormal drop in systolic blood pressure greater than 10mm Hg during inspiration)
- Neck vein distention
- CXR- “Water bottle” configuration
- Pericardial friction rub
 - Scratchy 3 component friction rub, best heard with the patient leaning forward

ECG findings show diffuse ST segment elevation except VR and V1 and PR segment depression

Commonly presents with sharp pain, aggravated by inspiration and relieved by sitting up and leaning forward

Dressler's Syndrome

- Autoimmune induced fibrinous pericarditis occurring anywhere between **2nd week and 2nd month** post MI
- Autoantibodies directed against damaged pericardial antigens
- Similar symptoms as acute pericarditis

Question

- A 58 year old man suffered a myocardial infarction involving the apex of the left ventricle. Six months later, an echocardiogram shows the development of a ventricular bulge that does not contract during systole. The patient subsequently suffers a massive stroke and suddenly dies. Which of the following would be an expected finding at autopsy?
- A. Calcific aortic stenosis
- B. Dilated cardiomyopathy
- C. Mitral valve prolapse
- D. mural thrombus
- E. Ventricular rupture

Question

- A 52 year old man with a history of diabetes mellitus and hypertension experiences **pain in his left arm and shoulder** for the past 2 hours. Over the next 6 hours he develops shortness of breath which continues for 2 days. On **day 3** he sees his doctor who notes a normal temperature, a pulse of 82 bpm, respirations of 18 bpm, and blood pressure of 160/100 mm Hg. Lab studies at this time show that **creatinine kinase serum levels are within normal** limits but **troponin I level is increased**. The patient is admitted to the hospital and continues to have dyspnea for the next 3 days. On **day 7 after the initial onset of shoulder** pain, the patient has a **cardiac arrest** and does not survive resuscitative efforts. An autopsy performed shows a large **transmural infarction** of the left anterior free wall with **rupture** and hemopericardium. Which statement is most likely true?

Answer Choices

- A. Myocardial infarction did not develop until day 5 or 6 after the episode of shoulder pain
- B. The normal CK level on day 3 excludes the possibility of myocardial infarction within the preceding 72 hours
- C. the patient had an acute myocardial infarction on the day he developed shoulder pain
- D. A CK-MB level would have detected an acute MI on day 3
- E. The patient had a second MI which caused a myocardial rupture on day 7

Summary Slide

- Myocardial infarction – definition
- Subendocardial v. Transmural
- Frequency and sites
- EKG changes
- Time sequence changes – see text, power point slide for details
 - 0-30 minutes reversible no real change
 - 30 min – 4 hrs “wavy fibers”
 - 4 – 12 hrs gross: focal mottling **HISTO: edema, start coag necrosis**
 - 12 – 24 hrs: gross: dark mottling **HISTO: coag necrosis, pyknosis, contraction bands, start PMNs**
 - 1 – 3 days: gross: yellow, tan center **HISTO: neutrophil infiltration**
 - 3 – 7 days: gross: hyperemic border, central yellow-tan **HISTO prominent macrophages, beginning granulation tissue**
 - 7 – 10 days: **HISTO: granulation tissue prominent**
 - 2 weeks: **HISTO: increased collagen formation**
 - 2 months – **HISTO: well formed scar**
- Laboratory tests/clinical correlation
- Complications – arrhythmias, pericarditis (acute fibrinous), pericarditis (Dressler syndrome), ventricular free wall rupture, false v. true aneurysm, papillary muscle rupture, ventricular septal rupture, contractile dysfunction